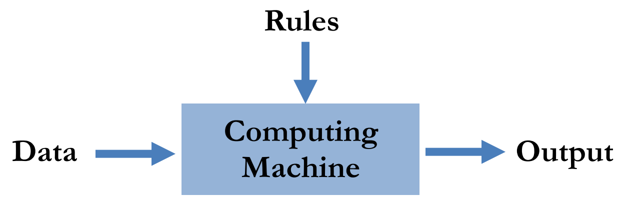
Introduction to Machine Learning

Chapter 1: **The Machine Learning Landscape**

Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, Gurelien Geron, 3e, O'Reilly Media, 2022.

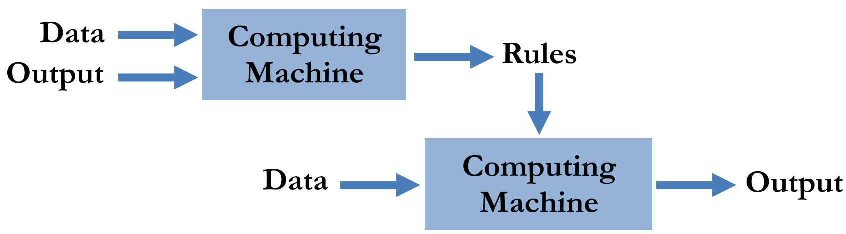
Traditional Approach

- You tell a computer what to do.



Machine Leaning Approach

- You show a computer how to do a task.
- Mimics how kids learn: we show them apples of different colors, of different shapes, from different angles, etc.



What is Machine Learning?

Definition 1

- Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed. (Arthur Samuel, 1959)
  - Samuel's checkers program

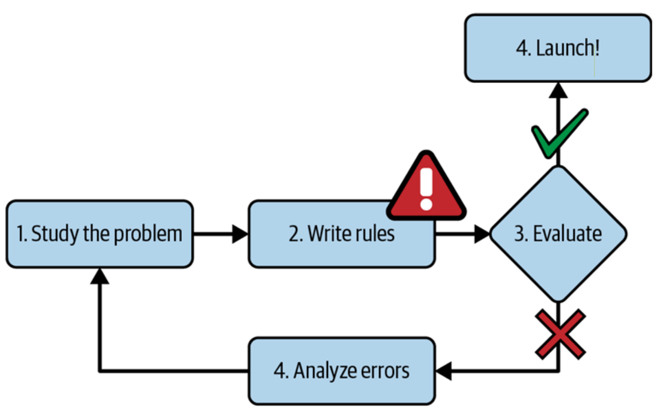
Definition 2:

- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E.
  (Tom M. Mitchell, 1997)
- The examples that the system uses to learn are called the training set. Each training example is called a training instance (or sample). This training set forms the *experience*.
- The part of a machine learning system that learns and makes predictions is called a model.

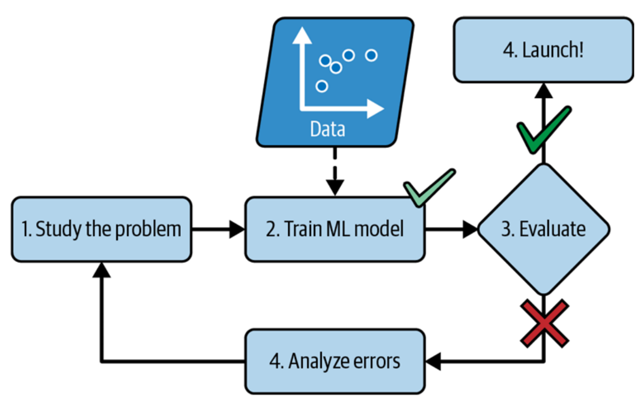
Why Use Machine Learning?

Machine learning is great for:

1. Problems for which existing solutions require a lot of hand-tuning or long lists of rules: one machine learning algorithm can often simplify code and perform better.



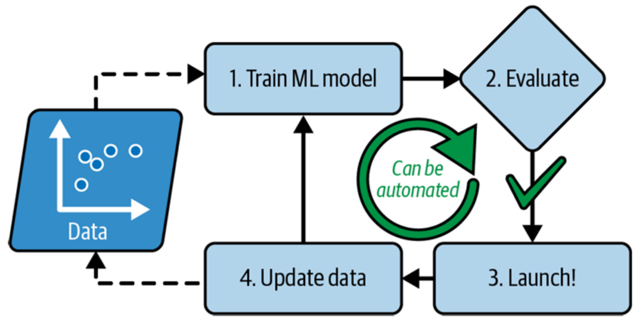
The traditional approach



The machine learning approach

2. Complex problems for which there is no good solution at all using a traditional approach: the best Machine Learning techniques can find a solution.
    - Another area where Machine Learning shines is for problems that either are too complex for traditional approaches or have no known algorithm. Tasks that fall into this category include multiple objects detection and localization, automatic speech recognition and other non-linear problems.

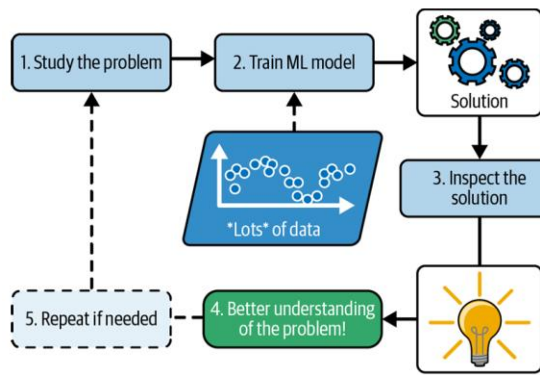
3. Fluctuating environments: a Machine Learning system can adapt to new data.
    - A machine learning algorithm can be made to automatically adapt to change.



Adapting to change

ML helps handle complex problems and big data
It can reveal patterns humans might not see
It can sometimes help us learn from data
This discovery of hidden patterns is called data mining

4. Getting insights about complex problems and large amounts of data.
    - Finally, Machine Learning can help humans learn.
    - ML algorithms can be inspected to see what they have learned (although for some algorithms this can be tricky).
    - Applying ML techniques to dig into large amounts of data can help discover patterns that were not immediately apparent. This is called data mining.



Machine learning can help humans learn

Machine Learning Everywhere!

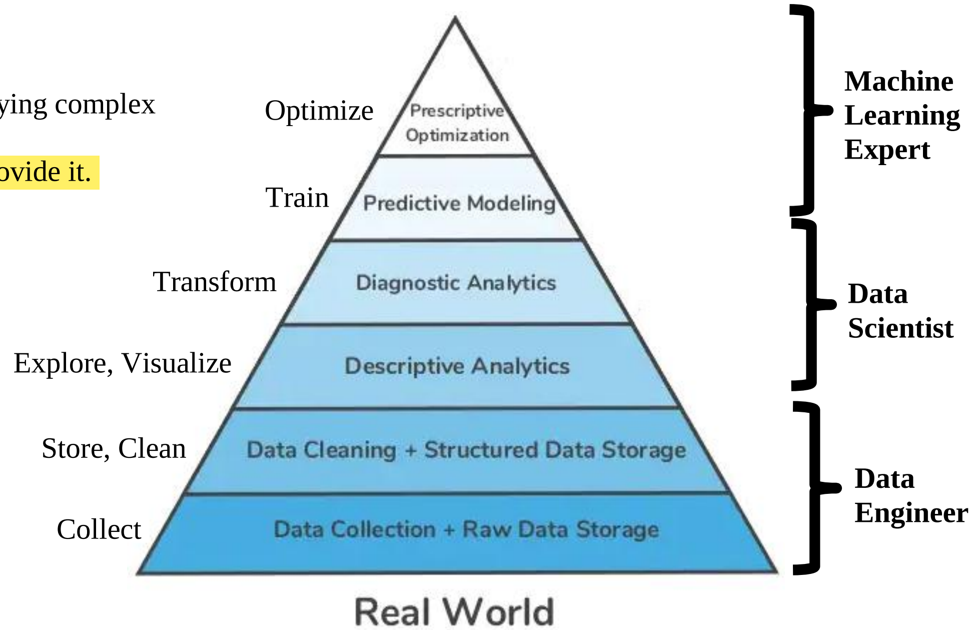
- Writing computer programs to do interesting stuff involving learning is a tough thing to do.
- It was realized that the only way to do this is by making the machine learn by itself.
- A computer observes some data, builds a model based on the data, and uses the model as both a hypothesis about the world and a piece of software that can solve problems.
- A diverse range of learning models are available, the key is to choose the right one for a problem at hand.
- Applications based on machine learning are ubiquitous now a days.
    - Search engines like Google or Bing, know how to rank web pages.
    - Facebook or Apple's photo application recognizes your friends in your pictures.
    - Email spam filters sort tons of spam and non-spam emails.
    - Medical records are analyzed to understand disease better.
    - Computational biology helps understand human genome better.
    - Analyzing images of products on a production line to automatically classify them.
    - Detecting tumors in brain scans.
    - Automatically classifying news articles.
    - Automatically flagging offensive comments on discussion forums.
    - Creating a chatbot or a personal assistant.
    - Forecasting your company’s revenue next year, based on many performance metrics.
    - Making your app react to voice commands.
    - Detecting credit card fraud.
    - Segmenting clients based on their purchases so that you can design a different marketing strategy for each segment.
    - Representing a complex, high-dimensional dataset in a clear and insightful diagram.
    - Recommending a product that a client may be interested in, based on past purchases.
    - Building an intelligent bot for a game.

Data Science

- Data mining + Computer Science = Data Science
- Make most of the data collected on every click, get useful insights using artificial intelligence and machine learning.
- Real world problems - clean, segment and visualize data; apply shallow and deep learning strategies to extract meaning from the data.

Hierarchy of needs

- Start with simple analysis before applying complex models.
- ML is only as good as the data you provide it.
- Quality of data is the key!



Addressing a Machine Learning Problem

- **Problem / Question Formulation**
  - Good model design depends on a well-defined question or hypothesis.
- **Data Collection**
  - Collecting and preparing relevant data is a crucial step in any machine learning project.
  - Most machine learning algorithms perform poorly without a well-prepared dataset.
  - Large volumes of data are often required for effective model training.
- **Data Preprocessing**
  - Choose appropriate formats and structures for storing data.
  - Data Cleaning: handle missing, incomplete, inconsistent, or inaccurate data.
- **Data Exploration and Visualization**
  - Verify that the data represents the real-world problem correctly.
  - Inspect and analyze data using statistical summaries and visualizations.
- **Feature Engineering**
  - Design informative features through feature extraction, transformation, and aggregation.
- **Model Training**
  - Train suitable algorithms such as linear regression, k-nearest neighbors, Naïve Bayes, decision trees, etc.
- **Model Evaluation**
  - Evaluate and compare models using appropriate performance metrics.
  - Select the best model for the given problem.

Main Challenges of Machine Learning

- Since the main task in ML is to select a learning algorithm and train it on data, two major problems can occur:
  - Bad algorithm (wrong choice or poorly suited method)
  - Bad data (insufficient, biased, or noisy data)
- **Insufficient Data:** ML models need large datasets; small sets cause poor accuracy.
- **Non-representative Data:** If data doesn't reflect real-world cases, predictions are biased.
- **Poor-Quality Data:** Noisy, incomplete, or error-filled data hides true patterns.
- **Irrelevant Features:** Too many useless features hurt learning; relevant ones improve it.
- **Wrong Algorithm Choice:** A poor algorithm may overfit or underfit.

Types of Machine Learning Systems

- There are so many different types of Machine Learning systems that it is useful to classify them in broad categories based on:
  - Whether or not they are trained with human supervision (supervised, unsupervised, semi supervised, and reinforcement learning)
  - Whether or not they can learn incrementally on the fly (online versus batch learning)
  - Whether they work by simply comparing new data points to known data points, or instead detect patterns in the training data and build a predictive model, much like scientists do (instance-based versus model-based learning)
- These criteria are not exclusive; you can combine them in any way you like. For example, a state-of-the-art spam filter may learn on the fly using a deep neural network model trained using examples of spam and ham; this makes it an online, modelbased, supervised learning system.

Supervision in Learning

- Machine Learning systems can be classified according to the amount and type of supervision they get during training.
- There are four major categories:
  - Supervised Learning
  - Unsupervised Learning
  - Semi-supervised Learning
  - Self-supervised Learning
  - Reinforcement Learning

Supervised Learning

- In supervised learning, the training data you feed to the algorithm includes the desired solutions, called labels.
- Given a data set, we already know what our correct output should look like, having the idea that there is a relationship between the input and the output.
- The agent observes input-output pairs and learns a function that maps from input to output.
- Supervised learning problems are categorized into regression and classification problems.

Regression

- In a regression problem, we try to predict results within a continuous output.
  - Map input variables to some continuous function.
  - The output can take any value.

Classification

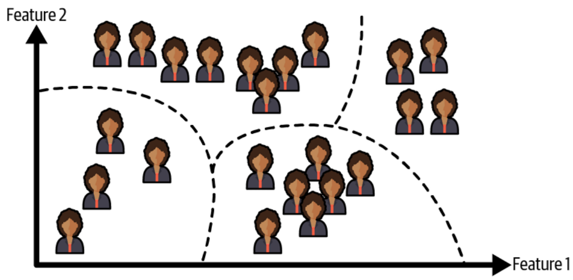
- In a classification problem, we try to predict results in a discrete output.
  - Map input variables into discrete categories.
  - The output can take only some discrete values.

Unsupervised Learning

- In unsupervised learning the dataset is a collection of unlabeled examples.
- Unsupervised learning allows us to approach problems with little or no idea what our results should look like.
- The agent learns patterns in the input without any explicit feedback.
- Unsupervised learning algorithms can be categorized into clustering, anomaly detection and novelty detection, dimensionality reduction, and association rule learning.

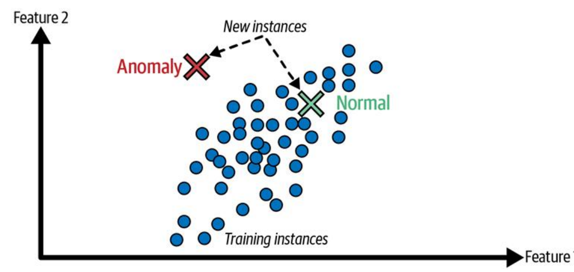
Clustering

- Clustering finds groups, having similar characteristics in unlabeled dataset.
  - More generic in nature.



Anomaly Detection and Novelty Detection

detects the anomaly



- The system is shown mostly normal instances during training, so the machine learning algorithm learns to recognize them and when it sees a new instance it can tell whether it looks like a normal one or whether it is likely an anomaly or novelty.
- Example: Detecting unusual credit card transactions to prevent fraud.

Dimensionality Reduction

- In dimensionality reduction, the goal is to simplify the data without losing too much information. One way to do this is to merge several correlated features into one.
- It is often a good idea to try to reduce the dimension of the training data using a dimensionality reduction algorithm before feeding it to another machine learning algorithm. It will run much faster, the data will take up less disk and memory space, and in some cases it may also perform better.
- Example: A car's mileage may be very correlated with its age, so the dimensionality reduction algorithm will merge them into one feature that represents the car's wear and tear. This is called feature extraction.
- Visualization algorithms also use dimensionality reductions to transform high-dimensional, unlabeled data into 2D or 3D representations while preserving data structure, making patterns and clusters easier to understand.

Association Rule Learning

- In association rule learning, the goal is to dig into large amounts of data and discover interesting relations between attributes.
- Example: Suppose you own a supermarket. Running an association rule on your sales logs may reveal that people who purchase eggs, also tend to buy bread. Thus you may want to place these items close to each other.

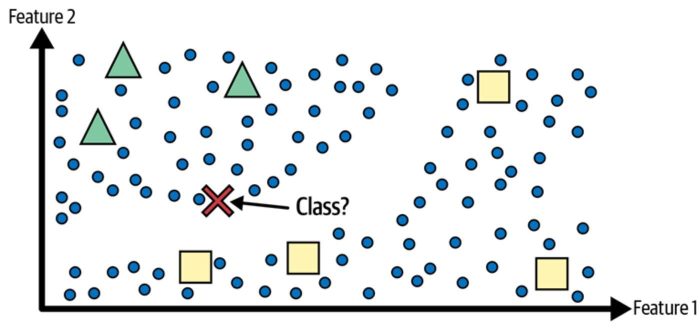
Practice Exercise

Of the following examples, which ones do you think are Unsupervised Learning problems as opposed to Supervised Learning problems? Also classify the sub-type.

1. Given data about the size of houses on the real estate market, try to predict their price.
2. Given data about house prices on a real estate market, predict if a house sells for more or less than the asking price.
3. Given a picture of a person, we have to predict their age on the basis of the given picture.
4. Given a patient with a tumor, we have to predict whether the tumor is malignant or benign.
5. You need to predict if people who purchase barbecue sauce and potato chips also tend to buy steak. Thus, you may want to place these items close to each other in your supermarket.
6. Take a collection of 1,000,000 different genes, and find a way to automatically group these genes into groups that are somehow similar or related by different variables, such as lifespan, location, roles, and so on.
7. On Google News (https://news.google.com), when you search a particular news it shows you other related news along with it.
8. Organize large computer clusters in a data center to figure out which machines tend to work together and if you can put those machines together, you can make your data center work more efficiently.
9. Social network analysis, where given knowledge about which friends you email the most or given your Facebook friends or your Google+ circles, can automatically identify which are cohesive groups of friends, also which are groups of people that all know each other.
10. Market segmentation, where using huge databases of customer information, you automatically discover market segments and automatically group your customers into different market segments so that you can automatically and more efficiently sell or market your different market segments together.
11. We need an algorithm to catch manufacturing defects in products.
12. We need an algorithm that can automatically remove outliers from a dataset before feeding it to another learning algorithm.
13. Suppose you are working on weather prediction, and use a learning algorithm to predict tomorrow's temperature (in degrees Centigrade/Fahrenheit).
14. Take a collection of 1000 essays written on the US Economy, and find a way to automatically group these essays into a small number of groups of essays that are somehow "similar" or "related".
15. Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Semi-supervised Learning

- The agent learns from both supervised and unsupervised ways.
- These algorithms deal with partially labeled training data, usually a lot of unlabeled data and a little bit of labeled data.
- Example: In Google Photos, once you upload all your family photos to the service, it automatically recognizes that the same person A shows up in photos 1, 5, and 11, while another person B shows up in photos 2, 5, and 7. This is the unsupervised part of the algorithm (clustering). Now all the system needs is for you to tell it who these people are. Just one or a few labels per person and it is able to name everyone in every photo, which is useful for searching photos.



Self-Supervised Learning

- Self-supervised learning is an approach where a model automatically generates labels from unlabeled data and is trained using supervised learning techniques.
- In other words, the data is unlabeled, but the supervision signal is created from the data itself.
- This allows the model to learn useful features from large unlabeled datasets. Later, these learned features can be fine-tuned on a smaller labeled dataset for a specific task. This is called transfer learning.
- Transfer learning is the process of transferring knowledge from one task to another, and it is especially important in modern machine learning with deep neural networks.

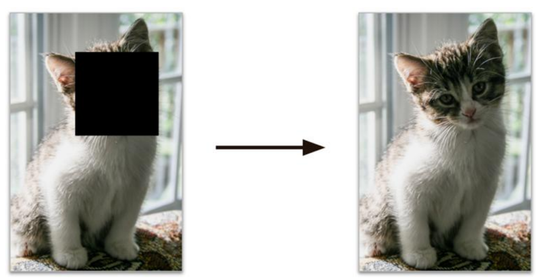
How It Works

1. Start with a large unlabeled dataset.
2. Create an artificial task:
   - Input = modified version of the data
   - Label = original data (or part of it)
3. Train the model to predict the missing or modified part.
4. Once trained, reuse and fine-tune the model for the real task.

- Self-supervised learning builds general knowledge from unlabeled data. Transfer learning leverages this knowledge for specific tasks with minimal labeled data, making modern AI practical and efficient.

Image Example

Self-Supervised Learning



- Take an image.
- Randomly mask part of it.
- Train a model to reconstruct the missing region.
- Input: masked image
  Label: original image
  The labels are automatically generated, not manually annotated.
- While learning to repair images, the model must understand shapes, textures, and object structures, which builds powerful visual representations.

Transfer Learning

- Final task is to classify pet species.
- After self-supervised pre-training, the model already understands visual features of pets.
- Replace the output layer and fine-tune using a small labeled dataset.
- Since the model already "knows" what cats and dogs look like, far fewer labeled examples are needed.

Reinforcement Learning

- The agent learns from a series of reinforcements: rewards and punishments.
- In general, a reinforcement learning agent is able to perceive and interpret its environment, take actions and learn through trial and error.
- It must then learn by itself what is the best strategy, called a policy, to get the most reward over time.
- A policy defines what action the agent should choose when it is in a given situation.

Batch Versus Online Learning

- Another criterion used to classify Machine Learning systems is whether or not the system can learn incrementally from a stream of incoming data.

Batch Learning

- Trains on all available data at once; cannot learn incrementally.
- Typically done offline, before deployment.
- After training, the system applies what it learned without further learning.
- To incorporate new data, the entire model must be retrained on both old and new data.
- Training is time- and resource-intensive, often done daily or weekly.
- Suitable for stable datasets, but not ideal for rapidly changing environments or resource-limited devices.

Online / Incremental Learning

- Trains sequentially, feeding data one instance or small batch at a time.
- Learning is fast, cheap, and adaptive, ideal for continuous data streams (e.g., stock prices, sensor data).
- Can handle very large datasets that cannot fit in memory (out-of-core learning).
- Once data is learned, it can often be discarded, saving storage.
- Sensitive to bad data, which can degrade performance; requires monitoring and safeguards.
- The model may gradually forget older knowledge when trained on new data continuously. This is called catastrophic forgetting. For example, you do not want the spam filter to forget all the previous spam words once new spam words are learned.

Instance-Based Versus Model-Based Learning

- One more way to categorize machine learning systems is by how they generalize.
- Most Machine Learning tasks are about making predictions.
- This means that given a number of training examples, the system needs to be able to generalize to examples it has never seen before.
- Having a good performance measure on the training data is good, but insufficient; the true goal is to perform well on new instances.
- There are two main approaches to generalization: instance-based learning and model-based learning.

Instance-based learning

- Instance based system learns the examples by heart, then generalizes to new cases by comparing them to the learned examples (or a subset of them), using a similarity measure.
- Also called non-parametric learning.

Model-based learning

- Model based system generalizes from a set of examples to build a model of these examples, then use that model to make predictions.
- Also called parametric learning.